

1
2 Personalized AI assistants could either promote or hinder the
3 emergence of collective intelligence

4
5 Jaeseung Kim¹ Sang Woo Park^{1,2,*}

6 **1** School of Biological Sciences, Seoul National University, Seoul, Korea

7 **2** Institute for Data Innovation in Science, Seoul National University, Seoul, Korea

8 *Corresponding author: sangwoopark@snu.ac.kr

9 **Abstract**

10 Advances in artificial intelligence (AI) have enabled the wide use of personalized
11 assistance for cognitive tasks, which enhanced individual performance on various
12 fields. However, the impact of AI assistance on collective intelligence remains poorly
13 understood. AI assistance may undermine the emergence of collective intelligence by
14 reducing the individual diversity or introducing structured biases into a population.
15 Here, we develop a human-AI collective prediction model mediated by social learning
16 based on incentive structure by extending the existing works on collective intelligence.
17 Our model uncovers that emergence of collective intelligence is depend on how human
18 and AI interact. When AI works as if it is omniscient and gives assistance regardless
19 of users' interests, collective intelligence will be promoted. However, when AI works
20 as a chat-bot and gives assistance only based on users' interests, collective intelligence
21 could be hindered due to diversity loss and overconfidence on AI. We showed that,
22 to promote collective intelligence, penalizing or incentivizing AI usage is not reliable.
23 Instead, elaborating more diversity-aware incentives could be the key to promote and
24 maintain the collective intelligence.

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning systems [10, 11]. Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences [12], evolutionary biology [13], and computer science [14]. Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes [15].

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks, including scientific discovery [16] and creative writing [17]. In particular, while the use of generative AI models, such as ChatGPT, has been rapidly increasing [18], the potential long-term impact of personal AI assistants on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise [19], and opinion aggregation [20]. Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

Our ability to predict the long-term impact of personal AI assistants on human CI is further hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of “tokens” used is often perceived as a proxy for productivity [26, 27]. On the other hand, many academic institutions appear to discourage, and even penalize, AI uses, reflecting concerns about plagiarism, misinformation, and limited independence [28, 29]. Such discrepancy raises a question about whether directly incentivizing or penalizing AI usage will benefit or harm human CI. These observations further highlight the importance of identifying a robust incentive structure that can promote CI in the presence of personal AI assistants. Such incentive structure should seek to retain the benefits of AIs and prevent potential risk, such as preserving the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, where each individual makes independent predictions for the outcome from partial observations of underlying causal factors [12, 13, 30]. In particular, we allow each individual to independently consult with personal AI assistants and revise their predictions accordingly. The resulting model allows us to investigate the robustness of human CIs under different assumptions about the AI model’s ability to make predictions as well as incentive structures that are associated with the prediction task. In doing so, we assess the impact of directly incentivizing or penalizing AI uses on CI and further suggest a balanced incentive structure that can maintain CI in the presence of AI.

66 Main

67 Model

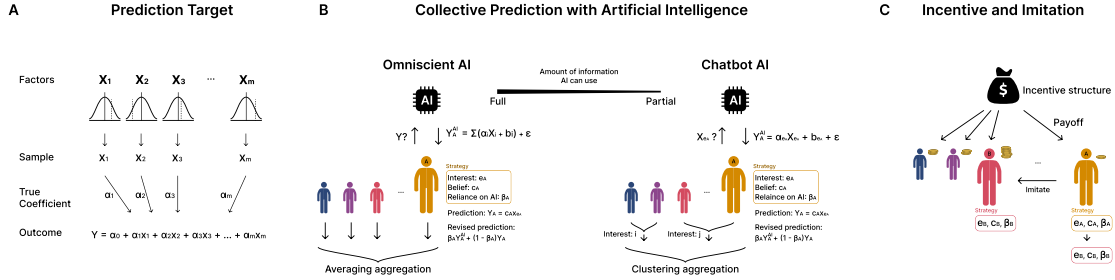


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of personal AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors. (B) Each player is characterized by three strategies: interest, belief, and reliance on AI. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

68 We begin by extending previously established models of CI, in which a group of
 69 players participates in a prediction task (Figure 1): each player makes an individual
 70 prediction based on partial observations of the underlying causal factors, and the
 71 group forms a collective prediction by aggregating these individual predictions [30,
 72 12, 13]. Specifically, the prediction target (Y) is modeled as a linear combination of
 73 M causal factors x_i , with fixed coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor
 74 is drawn from independent normal distribution (Figure 1A). Then, each individual
 75 A observes a single factor of their interest x_{e_A} and makes a prediction based on
 76 their own belief c_A : $Y_A = c_A x_{e_A}$ (Figure 1B). We assume each individual can then
 77 independently consult their AI assistants and revise their prediction based on their
 78 reliance on AI β_A : $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$ (Figure 1B). The revised predictions are
 79 then aggregated using two alternative aggregation rules, averaging and clustering, to
 80 form collective prediction.

81 To model AI, we assume that AI has already been trained on similar tasks and is
 82 able to make predictions based on partial or full observations of causal factors (Figure
 83 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based on
 84 its internal parameters, which can include factor-specific bias (b_i) and stochastic
 85 error ϵ . Here, we consider two extreme types of AI response, which differ in the
 86 amount of information AI uses when making predictions. In one extreme, the AI
 87 can observe all causal factors and use all information to make predictions, regardless

88 of the user’s interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works
 89 like a chatbot, where the AI predicts the outcome based only on the user’s interest:
 90 $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’
 91 and ‘Chatbot AI’, respectively.

92 *[JS: I added a new paragraph explicitly explaining the different aggregation rules*
 93 *used for each type of AI.]* The appropriate aggregation rule for collective prediction
 94 depends on the type of AI. With Omniscient AI, each revised prediction contains
 95 information about all causal factors and therefore represents an estimate of the full
 96 prediction target. With Chatbot AI, each revised prediction remains an estimate of
 97 the contribution of a single factor to the target. Based on this distinction, we use
 98 averaging aggregation for Omniscient AI, in which revised individual predictions are
 99 simply averaged to form the collective prediction. For Chatbot AI, we use cluster-
 100 ing aggregation, in which revised predictions are first averaged among individuals
 101 interested in the same factor, and then summed across factors.

102 Each generation consists of multiple round of predictions, and each player is re-
 103 warding a payoff according to the incentive structure (Figure 1C). For simplicity,
 104 we assume that a sufficiently large number of prediction rounds occurs within each
 105 generation, allowing us to use the expected payoff value rather than computing the
 106 payoffs for every round. We consider two incentive structures from previous work,
 107 “feedback” and “niche-expert”, which have been shown to promote near-perfect col-
 108 lective accuracy in the absence of AI [13]. The feedback structure gives higher payoff
 109 to individuals who can push the collective prediction toward the true target, regard-
 110 less of their individual predictive accuracy: $\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})$. In contrast, the
 111 niche-expert structure gives higher payoff to individuals who have both better pre-
 112 dictive accuracy and a minority interest: $\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2$. Here, ρ_{e_A} denotes
 113 the proportion of players interested in the same factor, thereby allowing the niche-
 114 expert structure to preserve interest diversity. We note collective intelligence fails to
 115 emerge under the combination of averaging aggregation with niche-expert structure
 116 in the absence of AI [13].

117 Through time, the group evolves by allowing each individual to imitate peers
 118 who received higher payoffs (Figure 1C). Specifically, after each generation, two in-
 119 dividuals are randomly sampled, and one imitates all three strategies (the factor of
 120 interest, factor belief, and reliance on AI) of the other with a probability determined
 121 by the difference in their expected payoffs (see the Simulations section in Methods).
 122 Individual predictions are aggregated into a collective prediction $\hat{Y}$ by simple averag-
 123 ing when using Omniscient AI and by clustering when using Chatbot AI. Collective
 124 accuracy is defined as the reduction in prediction error achieved by the collective
 125 prediction relative to a baseline prediction α_0 : $1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$. Hereafter, we say that
 126 CI emerges when the collective accuracy reaches a near perfect level.

127 This model now allows us to explore the impact of AI on CI depending on the
 128 mode of AI and the incentive structure. Here, we begin by comparing the emergence
 129 of CI with and without AI and subsequently investigate a how different incentive
 130 structures affect the emergence CI in the presence of AI.

131 AI can either promote or prevent the emergence of CI

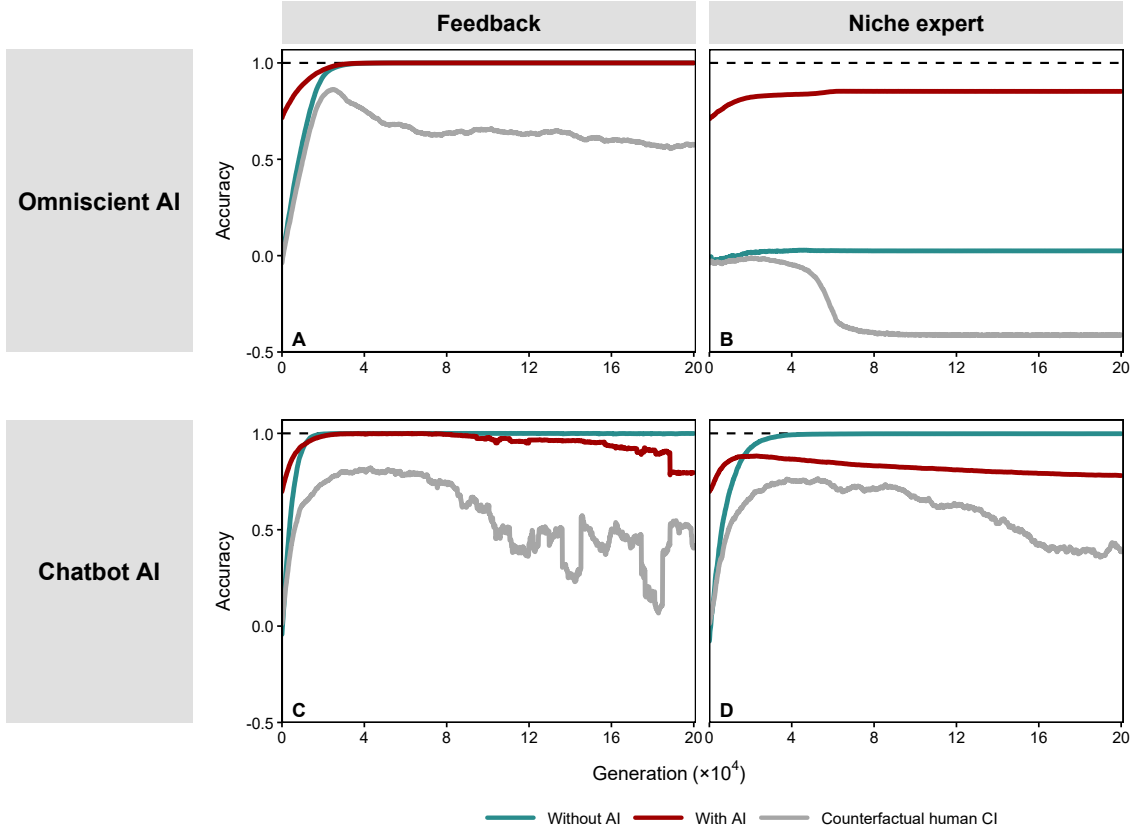


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by the Omniscient AI (A,B) and Chatbot AI (C,D). *[SWP: Maybe we should also name the averaging rule below AI...]* Here, feedback (A,C) and niche-expert (B,D) incentive structures are considered. Green lines represent trajectories of collective accuracy without AI assistance. Red lines represent trajectories of collective accuracy with AI assistance. Gray lines represent trajectories of counterfactual collective accuracy, computed from players' unrevised predictions, indicating the group's predictive accuracy if AI assistance were unavailable at the given generation. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

132 To understand the impact of AI, we first compare collective dynamics with and
 133 without AI. First, consistent with [13], simple imitation rules and incentive structures
 134 allow collective intelligence to emerge in the absence of AI (Fig. 2A,C,D). One
 135 exception is the use of simple averaging with niche-expert incentive structures, under
 136 which the collective accuracy of the population remains near zero *[SWP: because...]*

137 (Fig. 2B) [SWP: maybe need a supp figure].

138 We find that AI usage can have heterogeneous outcomes depending on its type.
139 For example, the use of Omniscient AI can either preserve CI under the feedback
140 incentive structure (Fig. 2A) or promote higher collective accuracy under the niche-
141 expert incentive structure (Fig. 2B). These results are robust across different levels
142 of AI accuracy and can be shown analytically by assuming a sufficiently large group
143 of players (Fig. S2) [SWP: probably need to swap S2 and S1 so that the current S2
144 appears first].

145 In contrast, the use of Chatbot AI prevents the emergence of CI, thereby resulting
146 in lower levels of collective accuracy (Fig. 2C, D). Under the feedback incentive
147 structure, the use of Chatbot AI leads to a loss of diversity in the population’s
148 interest, making it impossible to account for effect of some causal factors [SWP:
149 Supp]. This causes an increase in collective variance and stochastic decrease in
150 collective accuracy [SWP: Supp]. We note that the loss of interest diversity may
151 arise even in the absence of AI in finite populations [SWP: Supp], but the use of
152 Chatbot AI makes the group more vulnerable to such loss [SWP: Supp]. Under
153 the niche-expert incentive structure, the population becomes overly reliant on AI,
154 causing the collective prediction to inherit the AI’s bias (Fig. S1). This leads to
155 an increase in collective bias and a decrease in collective accuracy [SWP: Supp].
156 Interestingly, we find that the niche-expert structure can prevent the use of AI under
157 specific assumptions about the relationship between the true coefficient and AI bias
158 (Fig. S2). In this case, the population completely excludes AI assistance, allowing
159 the CI to emerge eventually. However, this phenomenon can substantially delay the
160 emergence of CI [SWP: Supp].

161 To investigate whether the group can retain its predictive ability when AI assis-
162 tance suddenly becomes unavailable at any given time, we quantify counterfactual
163 collective accuracy based on unrevised predictions. Across all scenarios, we find that
164 the counterfactual collective accuracy is lower than the collective accuracy of the
165 corresponding population.

166 In summary, an omniscient AI can promote CI, but AI operating with limited in-
167 formation may prevent the emergence CI. Moreover, reliance on AI can limit inherent
168 ability of the population to make accurate predictions on their own, raising concerns
169 about future AI usage. These findings further underscore the need for incentive
170 structures that can mitigate the potential risks associated with AI assistance.

171 Directly incentivizing or penalizing AI use can additionally 172 harm CI

173 Given the strong influence of AI on collective outcomes, a straightforward interven-
174 tion is to directly incentivize or penalize the use of AI. [SWP: maybe a sentence
175 about how this reflects real world scenarios.] To investigate whether such direct
176 incentives or penalties can help the emergence of CI, we modify the two incentive

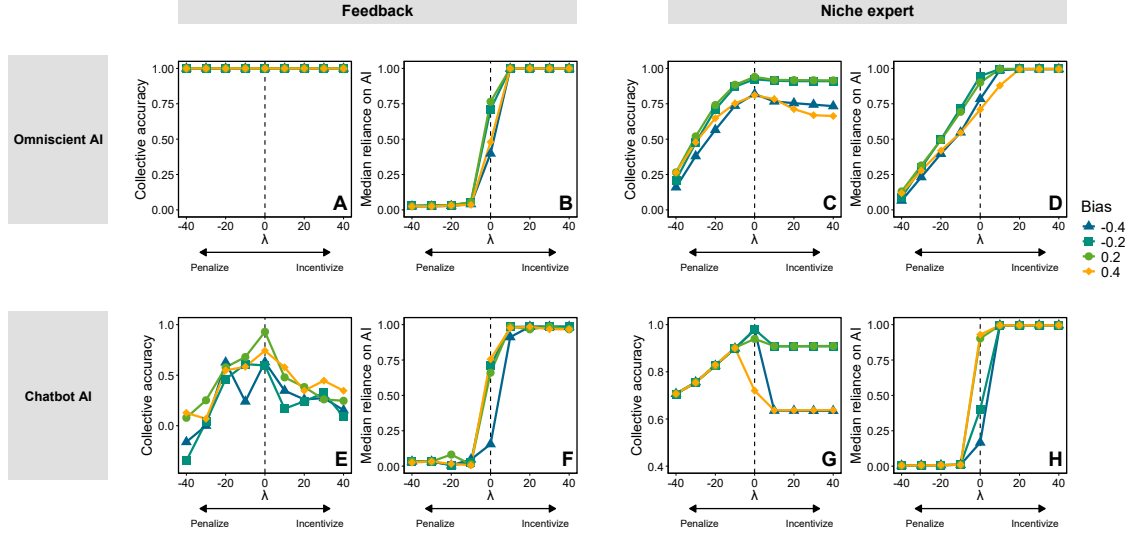


Figure 3: **Collective accuracy under the influence of direct intervention on AI use.** Collective accuracy and median reliance on AI under varying strengths of intervention (λ) across different assumptions about AI types (omniscient AI, A–D and chatbot AI, E–H) and incentive structures (Feedback, A,B,E,F, and Niche expert, C,D,G,H). Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. In each panel, the vertical dashed line indicates the original incentive structure ($\lambda = 0$). Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Collective accuracy (A,C,E,G) and median reliance on AI (B,D,F,H) are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

177 structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

178 where λ represent the strength of intervention ($\lambda > 0$) or penalty ($\lambda < 0$). [SWP:
179 maybe we need to call incentive pi earlier on...] [SWP: also good to define beta
180 again here as a reminder.]

181 Directly incentivizing or penalizing AI use can further reduce the collective ac-
182 curacy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback
183 structure; in the latter case, the population retains collective intelligence regardless
184 of the intervention strength (Fig. 3A). The reduction in collective accuracy arises be-
185 cause incentivizing or penalizing AI usage causes individuals to imitate others based
186 on their AI usage rather than their predictive accuracy. Thus, stronger interventions
187 can push the group to always ($\beta = 1$) or never ($\beta = 0$) rely on AI (Fig. 3B,D, F, H)
188 but these scenarios do not necessarily lead to better outcomes (Fig. 3D, F, H). When

189 AI bias is relatively large, a weak penalty can lead to higher collective accuracy, but
190 this represents a narrow parameter regime (Figure 3G).

191 Overall, these results indicate that simply incentivizing or penalizing the AI, in-
192 dependent of individuals' performance, use may exacerbate rather than resolve the
193 risk of AI on CI. Thus, we also considered an incentive structure where interven-
194 tions for AI usage affects the payoffs multiplicatively, thus giving higher payoffs to
195 individuals who rely more (or less) AI and makes better prediction. [SWP: Supp.]

196 **Balanced incentives mitigate AI-induced failures of CI**

197 Neither the original incentive structures nor simple interventions targeting AI use are
198 sufficient to promote CI. What incentive structure, then, would work in the presence
199 of AI? Here, we propose a “balanced” incentive structure, modeled as the average of
200 the feedback and niche expert incentives.

201 The core idea behind the balanced structure is to maintain interest diversity and
202 prevent collective bias. On one hand, the feedback incentive structure is prone to
203 the loss of interest diversity but can prevent collective bias by incentivizing players
204 whose predictions lie opposite to the collective prediction. On the other hand, the
205 niche-expert structure can maintain interest diversity because it directly incentivizes
206 minority interests. However the niche-expert structure cannot prevent collective bias
207 because, within the same factor of interest, individuals are rewarded only according
208 to their individual predictive performance, which in turn homogenize the predictions.

209 Therefore, when both incentive structures are combined, the population achieves
210 CI by maintaining both interest diversity and limited collective bias (Fig. 4, S3).
211 Compare to the feedback and niche-expert incentive structures, individuals are both
212 evenly distributed through factors (Fig. 4C–E), and their beliefs lie closer to true
213 values (Fig. 4F–H). Moreover, the balanced incentive structure allows for higher lev-
214 els of counterfactual collective accuracy compared to pure feedback and niche-expert
215 scenarios. These patterns are driven by the emergent property of the balanced struc-
216 ture, in which group increases its reliance on AI as the AI becomes more accurate,
217 demonstrating appropriately calibrated AI use.

218 [SWP: *may need some robustness check rather than a case study...*]

219 **Discussion**

220 The ongoing race towards the development of “better” AIs continues to pose ques-
221 tions about the societal benefits and risks of AI uses. Here, we used mathematical
222 models of evolutionary dynamics to investigate how AI uses and incentive structures
223 jointly affect the emergence of CI, which is defined as the population’s ability to make
224 accurate predictions. We find that an “omniscient” AI model, which incorporates in-
225 formation about all relevant causal factors, can improve collective accuracy, whereas
226 a “chatbot” AI model, whose outcome depends on the user’s interest, can impair it.
227 These contrasting outcomes arise because AI assistance affects not only individual

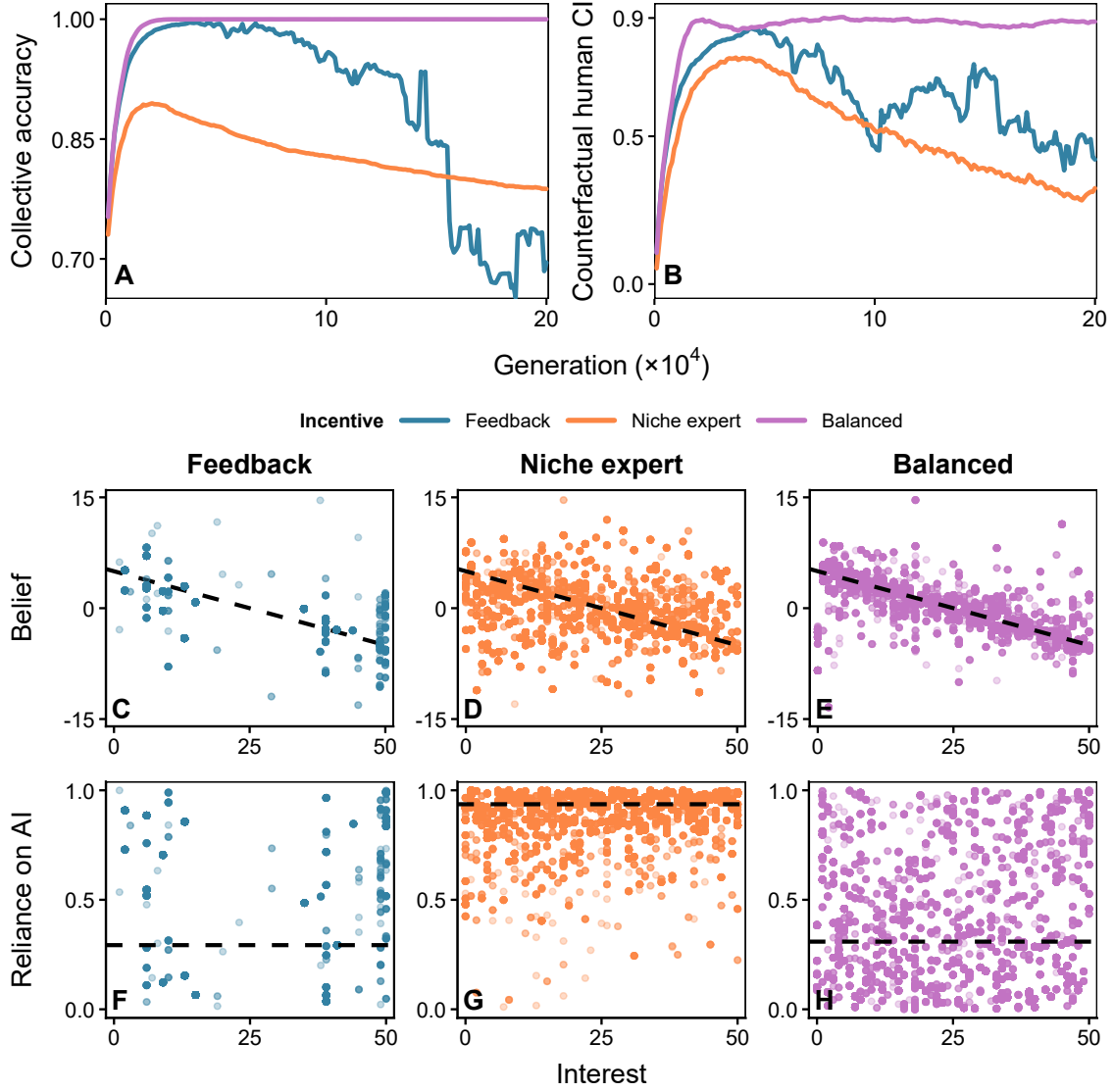


Figure 4: **Balanced structure can promote collective intelligence** Trajectories of collective accuracy (A) and counterfactual human CI (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. *[SWP: A and B should have the same y axes scaling.]* Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations. Each point represents an individual. Dashed lines in panels C-E represent the true coefficients (α_i) of the prediction target. Dashed horizontal lines in panels F-H indicate the median reliance on AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

228 predictions, but also the population’s interest diversity, beliefs, and reliance on AI.
229 We also show that directly incentivizing or penalizing AI uses, independent of in-
230 dividual’s performances, can further limit the collective accuracy of the population.
231 Instead, a balanced incentive structure can maintain CI by promoting diversity and
232 limiting bias.

233 Our analysis reveals that AI can prevent the emergence of CI through two dif-
234 ferent mechanisms. First, AI use can cause individuals to concentrate on a subset of
235 the factors that determine the prediction target, thereby reducing diversity in their
236 interests. This result is broadly consistent with recent empirical studies showing that
237 AI assistance can improve individual performance but limit the diversity of outputs,
238 in domains ranging from scientific research [31] to creative writing [17]. Second, over-
239 reliance on AI can cause the population to inherit inherent biases in AI’s predictions,
240 causing the prediction accuracy to decrease eventually. This mechanism can prevent
241 the emergence of CI even when the interest diversity is maintained because biases in
242 AI predictions can be amplified when individual predictions are aggregated. Thus,
243 preserving the diversity of ideas will not be sufficient to promote CI. Instead, AI
244 uses need to be calibrated to prevent systematic AI biases from becoming collective
245 biases.

246 Our analyses of counterfactual collective accuracy also raises an important con-
247 cern for future AI uses: while AI uses may improve the apparent performance of
248 individuals and, in some cases, the population as a whole, it can simultaneously re-
249 duce the population’s ability to make accurate predictions without AI. These results
250 echo findings from a recent field experiment, which demonstrated that students, who
251 seemingly performed better in solving mathematics problems under AI assistance,
252 performed worse on their own than those who did not have access [32]. These findings
253 highlight the need to evaluate AI tools not only by the performance gains they pro-
254 vide during use, but also by whether individuals and populations retain the ability to
255 perform without them. Making this distinction clear will be particularly important
256 for education, where AI may help students complete a task without helping them
257 learn how to do it on their own.

258 As generative AIs and large language models are already generating substantial
259 benefits, many organizations, both industrial and academic, are increasingly seeking
260 to improve their collective performance by regulating AI uses. Such regulations may
261 take direct forms, including AI-use policies or token-based leaderboards, or may op-
262 erate indirectly through other incentives, such as productivity rewards or the social
263 stigma associated with AI use. However, our model suggests that directly incentiviz-
264 ing or penalizing AI use is unlikely to induce appropriately calibrated reliance on
265 AI or to promote CI. Incentivizing AI use could lead to overreliance, causing groups
266 to propagate AI biases, whereas penalizing AI use could lead to underreliance, pre-
267 venting groups from using AI even when it performs well in the relevant domain.
268 Our result instead shows that appropriate AI use and accurate CI can be promoted
269 by focusing on the traditional properties of groups that have been emphasized for
270 decades: maintaining diversity and preventing collective bias, rather than on AI use

271 itself.

272 [SWP: *need to be edited*:] There are several limitations to our analyses. First,
273 the model considers only social learning among humans and treats the influence of
274 AI as unidirectional. In reality, humans can learn from AI, while AI systems can
275 also be further trained on human-generated data. Such bidirectional adaptation may
276 further reduce group performance by increasing the similarity between human and
277 AI predictions and thereby weakening their complementarity. Empirical studies of
278 human–AI collaboration likewise suggest that stronger correlations between human
279 and AI predictions can diminish the benefits of collaboration. Second, we model AI
280 errors using factor-specific additive biases and random error. Previous studies of AI
281 bias and error suggest that additive biases may arise from skewed training data, or
282 systematic calibration. However, AI errors may also exhibit more complex structures,
283 including biased estimates of factor effects and correlations across prediction errors.
284 Further empirical research is therefore needed to characterize the structure of AI
285 bias and error, together with modeling approaches that can incorporate these more
286 complex patterns. Finally, to retain the tractability of established CI models, we
287 assume that the prediction target is a linear combination of causal factors, which does
288 not hold for many real-world tasks. In particular, many machine learning methods
289 used in AI systems are capable of capturing nonlinear relationships. Future research
290 should therefore investigate how nonlinear relationships between causal factors and
291 prediction targets affect the interaction between AI assistance and the emergence of
292 CI.

293 Methods

294 Prediction target

295 Following previous models of CI, we model the prediction target as a linear combi-
296 nation of M independent causal factors:

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (2)$$

297 where the fixed coefficient α_i determines the magnitude and direction of the
298 effect of factor i , and α_0 is a global intercept. Each causal factor X_i is modeled as an
299 independent Gaussian random variable with mean zero and standard deviation σ_i .

300 Players

301 We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a novel
302 prediction task. In the absence of AI assistance, the strategy of player A is char-
303 acterized by two variables: interest e_A and belief c_A . Player A observes the causal
304 factor X_{e_A} and predicts the target by multiplying the observed factor value by their
305 belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

where e_A can take an integer value from 0 to M . When $e_A = 0$, the player does not observe any causal factor and instead makes a constant prediction based solely on their belief. We refer to this as a stubborn strategy because the player's prediction does not respond to the realized values of the causal factors. For notational convenience, we assume that $X_0 = 1$.

When the group is assisted by AI, each player's strategy additionally includes reliance on AI, denoted by β_A . We assume that each player consults the AI independently and revises their prediction by taking a weighted average of their own prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

AI models

We assume that the AI model has been trained on similar tasks and is therefore capable of making predictions based on causal factors. However, because the model is imperfect, its outputs are subject to systematic additive bias and stochastic error. The additive bias associated with factor i is denoted by b_i , and the stochastic error is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its biases therefore remain fixed throughout the simulation. Although the bias is indexed by factor, all b_i , including b_0 , are assigned the same value in the main analyses for simplicity. The model formulation nevertheless allows biases to differ among factors.

We assume that the stochastic error follows a Gaussian distribution with mean zero and standard deviation τ . Each time a player consults the AI, an independent error is generated. Errors are therefore independent across players and across repeated consultations by the same player. Consequently, stochastic AI errors can be attenuated when independently generated predictions are aggregated across players. We therefore held τ constant and varied the magnitude of the additive bias to manipulate the level of AI accuracy across analyses.

We consider two extreme types of AI that differ in the amount of information available for prediction. The first is termed the Omniscient AI, which can observe all M causal factors and incorporate them into its prediction regardless of the user's interest. The prediction generated by the Omniscient AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

The second is termed the Chatbot AI, which can use only the single causal factor X_{e_A} associated with the player's interest. The prediction generated by the Chatbot

AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

Aggregation rules

The revised individual predictions are aggregated to form a collective prediction. We use two aggregation rules proposed in previous research, depending on the type of AI used by the group.

When the group is assisted by the Omniscient AI, the AI component of each revised prediction contains information about all M factors, even though the individual's own prediction is based on only one factor. Each revised prediction can therefore be treated as an estimate of the entire target. Accordingly, we calculate the collective prediction by taking the simple average of the revised individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \dots + Y_n^{\text{re}}}{N}. \quad (7)$$

When the group is assisted by the Chatbot AI, both the player's initial prediction and the AI's prediction are based on the same causal factor. Each revised prediction therefore contains information about only one factor and can be treated as an estimate of that factor's contribution to the target. We consequently use clustering aggregation.

Clustering aggregation is performed by grouping individuals according to their interests and calculating the mean prediction within each group. The collective prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i: n_i > 0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number of players in that set.

When the group is not assisted by AI, collective predictions are computed from the unrevised individual predictions throughout the simulation. Both simple averaging and clustering can be used in the absence of AI. For fair comparison, we use the same aggregation rule as in the corresponding AI-assisted condition.

Incentive structures

Players receive payoffs determined by an incentive structure based on their strategies. We consider five incentive structures: feedback structure, niche-expert structure, direct intervention on AI use of these two structures, and a balanced structure.

The feedback structure does not directly reward individuals according to their own prediction accuracy. Instead, it rewards predictions that are aligned with the

371 residual error of the collective prediction and therefore have the potential to move
 372 the collective prediction toward the true value. Given player A 's revised prediction
 373 Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

374 The niche-expert structure assigns higher payoffs to individuals who make accu-
 375 rate predictions while having an underrepresented interest. Specifically, the squared
 376 prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players
 377 interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

378 Previous research has shown that these two incentive structures can produce
 379 nearly perfect CI through social learning, except when the niche-expert incentive is
 380 combined with simple averaging. They are therefore used as baseline structures for
 381 examining how AI assistance affects the emergence of CI.

382 To examine the effects of directly encouraging or discouraging AI use, we mod-
 383 ify the feedback and niche-expert structures by adding a term proportional to the
 384 player's reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A. \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

385 Here, λ is a weighting coefficient representing the strength of the direct incentive
 386 or penalty applied to AI use. A positive value of λ rewards AI reliance, whereas a
 387 negative value penalizes it. When $\lambda = 0$, no direct incentive or penalty is applied,
 388 and the payoff functions reduce to the original feedback and niche-expert structures.

389 The balanced structure is defined as the arithmetic mean of the feedback and
 390 niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

391 Simulations

392 All simulations were conducted using an agent-based model with 50 causal factors
 393 ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction
 394 target were sampled at the beginning of the simulations and subsequently held fixed.

395 The coefficients α_i were sampled from a uniform distribution on $[-5, 5]$, except in
 396 the simulation presented in Figure 4, in which α_i was set to $5 - 0.2i$ for visualization.
 397 The standard deviations σ_i were sampled from a uniform distribution on $[0, 3]$.

398 Players' initial strategies were also sampled from predefined probability distri-
 399 butions. The distribution of individual belief c_A depended on the aggregation rule.

Under simple averaging, each factor’s contribution is scaled by the proportion of players interested in that factor. We therefore sampled beliefs from a Gaussian distribution with mean zero and standard deviation 100, following the scaling used in previous work [13]. Under clustering aggregation, beliefs were sampled from a Gaussian distribution with mean zero and standard deviation 5. Because predictions are averaged separately within each interest group and then summed, no additional scaling of c_A is required. The initial interest of each player was sampled uniformly from the integers between 0 and 50, and reliance on AI was sampled from a uniform distribution on $[0, 1]$.

We assumed that prediction events occurred on a substantially faster timescale than social learning. At each generation, we therefore calculated the expected payoff $\mathbb{E}[\pi_A]$ of each player analytically rather than explicitly sampling prediction events. The expectation was taken over the distributions of the causal factors $X_1, \dots, X_M$ and the stochastic AI error ϵ , conditional on the current population strategies and the fixed target parameters. Closed form expression of expected payoffs were derived for all incentive structures and are presented in the Supplementary Methods.

Each generation consisted of an expected payoff calculation followed by one opportunity for pairwise social learning. Two distinct players, A and B , were randomly sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

where s represents the strength of social learning and was set to 50 in all simulations. We introduced a mutation rate μ , such that with probability μ , player A did not imitate player B but instead redrew all strategy components from the same distributions used to initialize the population. The mutation rate was set to zero in all simulations except those presented in Supplementary Figure 4.

Unless otherwise specified, each simulation was run for 200,000 generations. When a result was represented by a single summary value rather than as a temporal trajectory, the metric was averaged over the final 10,000 generations.

To evaluate the dynamics of collective prediction, we defined six summary metrics: collective accuracy, collective bias, collective variance, counterfactual collective accuracy, interest diversity, and median reliance on AI. At each population state, expectations and variances were evaluated analytically over the distributions of the causal factors and AI errors rather than estimated from sampled prediction events. Collective bias and collective variance constitute the bias–variance decomposition of the group level collective prediction error.

Collective accuracy was normalized relative to the baseline prediction α_0 . A negative value indicates that the collective prediction performs worse than the baseline. Counterfactual collective accuracy was evaluated by setting $\beta_A = 0$ while retaining the evolved interests and beliefs of the AI-assisted population. In calculating interest diversity, we used the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

439 **Data and code availability**

440 All data are stored in a publicly available GitHub repository ([https://github.com/](https://github.com/parklab-snu/collective-intelligence-with-AI)
441 [parklab-snu/collective-intelligence-with-AI](https://github.com/parklab-snu/collective-intelligence-with-AI)).

442 **Acknowledgements**

443 S.W.P. was supported by the New Faculty Startup Fund from Seoul National Univer-
444 sity, and the Global-LAMP Program of the National Research Foundation of Korea
445 (NRF) grant funded by the Ministry of Education (No. RS-2023-00301976).

446 **Contributions**

447 Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology,
448 Software, Visualization, Writing - Original Draft Preparation, Writing - Review &
449 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
450 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

451 **Competing Interests**

452 The authors declare no competing interests.

453 Supplementary Text

454 1 Omniscient AI

455 Omniscient AI assumes that AI can incorporate all causal factors regardless of the
456 player's interest.

457 1.1 Model

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (S1)$$

458 For simplicity, prediction of Omniscient AI can be written as:

$$Y^{AI} = Y + B + \epsilon \quad (S2)$$

459 where Y stands for true prediction target ($Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M$) and
460 B is sum of all fixed biases ($B = b_0 + b_1 + \cdots + b_M$), ϵ follows normal distribution
461 with zero mean, τ standard deviation with no correlation within or between players.
462 Player A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (S3)$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (S4)$$

464 where c_A stands for player A's belief on the factor, e_A stands for the interest, and
465 β_A stands for reliance on AI.

466 1.2 Aggregation

467 Because Y_A^{AI} is subject to all M factors, we use simple averaging aggregation for
468 collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \frac{1}{N} \sum_{j=1}^N Y_j^{re} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j Y^{AI} + (1 - \beta_j) Y_j \} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j (Y + B + \epsilon) + (1 - \beta_j) c_j X_{e_j} \} \end{aligned}$$

469 Reparameterization for simple derivation and interpretability.

$$\bar{\beta} := \frac{1}{N} \sum_{j=1}^N \beta_j \quad (S5)$$

470

$$\bar{\eta} := \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (\text{S6})$$

471

$$\gamma_i := \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (\text{S7})$$

472 $\bar{\beta}$ can be interpreted as mean reliance on AI, γ stands for the coefficient for the factor
 473 i which is solely based on players' belief.

474 By this reparameterization, we can write $\hat{Y}^{re}$ as follows.

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (\text{S8})$$

475 1.3 Expected payoffs

476 Feedback

$$\delta_0 := \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0) \quad (\text{S9})$$

477

$$\delta_i := \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0) \quad (\text{S10})$$

478

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S11})$$

479 See equation Eq. (S6) for the $\bar{\eta}$.

480 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \\ &= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \end{aligned}$$

481 **Niche-expert**

482 If $e_A = 0$, player does not observe causal factor and their unrevised prediction is same
 483 with their belief c_A . Such that, the expected payoff is

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \\
 &= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

484 Else,

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \\
 &= -\mathbb{E}\left[\rho_{e_A}\left\{(\beta_A - 1)\alpha_0 + \beta_A B + (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B + \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

485 **Incentivizing or penalizing AI use**

486 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
 487 payoff as

$$\mathbb{E}(\pi_{A, Feedback}) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A] = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] + \lambda\beta_A. \quad (S12)$$

$$\mathbb{E}(\pi_{A, Niche-expert}) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A] = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] + \lambda\beta_A. \quad (S13)$$

488 **1.4 Expected accuracy**

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (S14)$$

489

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S15})$$

490

$$\begin{aligned} R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\ &= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \end{aligned}$$

491 2 Chatbot AI

492 2.1 Model

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon \quad (\text{S16})$$

493 b_{e_A} stands for factor specific additive bias, ϵ follows normal distribution with zero
 494 mean, τ standard deviation with no correlation within or between players. Player
 495 A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S17})$$

496

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S18})$$

497 where c_A stands for player A's belief on the factor, e_A stands for the interest, and
 498 β_A stands for reliance on AI.

499 2.2 Aggregation

500 For Chatbot AI, clustering aggregation is used for collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A Y_A^{AI} + (1 - \beta_A) Y_A\} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon] \end{aligned}$$

501 Reparameterize for the simplicity.

$$\tilde{c}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \quad (\text{S19})$$

502

$$\tilde{B}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A} \quad (\text{S20})$$

503

$$\bar{B} := \sum_{i=0}^M \tilde{B}_i \quad (\text{S21})$$

504

$$\tilde{\eta}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon \quad (\text{S22})$$

505

$$\bar{\eta} := \sum_{i=0}^M \tilde{\eta}_i \quad (\text{S23})$$

506 2.3 Expected payoffs

507 Feedback

508

$$\delta_0 := \alpha_0 - \tilde{c}_0 - \bar{B} \quad (\text{S24})$$

509

$$\delta_i := \alpha_i - \tilde{c}_i, \quad (i \neq 0) \quad (\text{S25})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S26})$$

510 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \end{aligned}$$

511 Niche-expert

512 If $e_A = 0$,

$$\begin{aligned} \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\ &= -\mathbb{E} \left[\rho_0 \left[\{\beta_A \alpha_0 + (1 - \beta_A) c_A\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \\ &= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0) \beta_A b_0 + \beta_A^2 b_0^2 \right. \\ &\quad \left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \end{aligned}$$

513 Else,

$$\begin{aligned}
\mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{\text{re}} - Y)^2] \\
&= -\mathbb{E}\left[\rho_{e_A}\left[\{\beta_A\alpha_{e_A} + (1 - \beta_A)c_A\}X_{e_A} + \beta_A b_{e_A} + \beta_A\epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i\right]^2\right] \\
&= -\rho_{e_A}\left\{(1 - \beta_A)^2(c_A - \alpha_{e_A})^2\sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2\right. \\
&\quad \left.+ \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0\beta_A b_{e_A} + \beta_A^2 \tau^2\right\}
\end{aligned}$$

514 Incentivizing or penalizing AI use

515 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
516 payoff as

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] + \lambda\beta_A. \quad (\text{S27})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right] + \lambda\beta_A. \quad (\text{S28})$$

517 Balanced

518 Expected payoff of the balanced structure can also be driven by arithmetic mean of
519 expected payoff of feedback and niche-expert.

$$\mathbb{E}(\pi_A) = \mathbb{E}\left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}\right] = \frac{\mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] - \mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right]}{2} \quad (\text{S29})$$

520 2.4 Expected accuracy

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S30})$$

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S31})$$

$$\begin{aligned}
R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\
&= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2}
\end{aligned}$$

523 3 Evolutionary dynamics

524 For sufficiently large population ($N \rightarrow \infty$), we can approximate the deterministic
525 evolutionary dynamics using the replicator equation. Let λ denote a strategy, p_λ its
526 frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff. The replicator equation is

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S32})$$

527 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$. Therefore, at an internal stationary state, all strategies
528 with positive frequency have the same expected payoff. A boundary state can also
529 be stationary because a strategy with exactly zero frequency cannot increase under
530 the replicator equation. Thus, stability of a boundary state needs to be considered
531 separately by examining invasion by a rare strategy.

532 3.1 Omniscient AI

533 3.1.1 Feedback

534 When the population size approaches infinity, the effect of stochastic error on the
535 collective prediction vanishes ($\bar{\eta} \rightarrow 0$) and the effect of stochastic error in the ex-
536 pected payoff, $-\beta^2 \tau^2 / N$, also vanishes. Following prior research [13], we define the
537 Lyapunov function as the expected error of collective prediction

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S33})$$

538 The function V is non-negative and equals zero when the collective prediction is
539 identical to the target. To examine how V changes under the evolutionary dynamics,
540 we take its time derivative:

$$\begin{aligned}
\frac{dV}{dt} &= -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \\
&= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \\
&= -2 \left[\left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right].
\end{aligned}$$

541 For notational convenience, we define

$$K = (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S34})$$

542 Although K changes with the population state, it is common to all strategies at a
543 given state and can therefore be treated as a constant with respect to the strategy
544 index λ . We also define

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S35})$$

545 By the definitions of $\bar{\beta}$ and γ_i and the replicator equation,

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S36})$$

546 Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta) c G_e, u). \quad (\text{S37})$$

547 Therefore,

$$\begin{aligned}
\frac{dV}{dt} &= -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta) c G_e, u) \} \\
&= -2 \text{Cov}(K\beta + (1 - \beta) c G_e, u).
\end{aligned}$$

548 For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy
549 λ is

$$u_{\lambda} = K \beta_{\lambda} + (1 - \beta_{\lambda}) c_{\lambda} G_{e_{\lambda}}. \quad (\text{S38})$$

550 Which is same with Thus,

$$\frac{dV}{dt} = -2 \text{Cov}(u, u) = -2 \text{Var}(u) \leq 0. \quad (\text{S39})$$

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\text{Var}(u) = 0$. This condition corresponds to equal expected payoffs among the strategies present in the population. A boundary state in which the entire population consists of a single strategy can also satisfy $\text{Var}(u) = 0$ even when $V > 0$. However, such a state is not evolutionarily stable under the feedback structure. If the collective prediction differs from the target Y , a nonzero residual remains. By the definition of the feedback payoff, a novel strategy whose prediction aligns with the direction of this residual obtains a higher expected payoff than the resident strategy and can therefore invade the population. Thus, an imperfect boundary equilibrium with $V > 0$ is unstable to the introduction of novel predictions.

Consequently, the evolutionary dynamics continue to reduce the collective prediction error until the population reaches the stable state with zero residual, $V = 0$. At this state, the collective prediction equals the target, $\hat{Y}^{re} = Y$, and the collective accuracy reaches $R = 1$.

3.1.2 Niche-expert

Unlike the stochastic error in the collective prediction, stochastic error in the niche-expert payoff is evaluated at the individual level and does not vanish as population size increases. Therefore, we retain $\tau > 0$ in the following derivation.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S40})$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its unconstrained stationary point is

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S41})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

Let L_e denote the minimum expected individual error, $\mathbb{E}[(Y_A^{re} - Y)^2]$, within interest e . For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S42})$$

584 If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating
 585 the expected squared prediction error at the corresponding boundary.

586 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S43})$$

587 Substituting this into the expected payoff gives a strictly concave function of β ,
 588 with the unique optimum

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S44})$$

589 Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S45})$$

590 and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S46})$$

591 We denote the optimal strategy, or the payoff-equivalent strategy class when
 592 $\beta_e^* = 1$, by S_e . A state containing a suboptimal strategy within an interest can
 593 be stationary if the optimal strategy is absent, but it is unstable against invasion
 594 because the optimal strategy has a strictly higher payoff at the same ρ_e . Therefore,
 595 a stable state can contain only S_e within each represented interest.

596 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
 597 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S47})$$

598 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator
 599 equation at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \quad (\text{S48})$$

600 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S49})$$

601 This internal stationary state is locally stable. At equilibrium, all represented
 602 interests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
 603 interest increases slightly above its equilibrium value, its payoff decreases because
 604 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of
 605 an interest decreases below its equilibrium value, its payoff increases. The replicator
 606 dynamics therefore generate a restoring force that reduces small deviations in interest
 607 frequencies and drives the population back toward the internal stationary state.

608 If one or more interests are absent, the resulting boundary state can be stationary
 609 under the replicator equation, but it is unstable against invasion. The payoff of a rare
 610 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
 611 population is negative when $L_e > 0$, so the rare interest increases when introduced
 612 at sufficiently small frequency.

613 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
 614 prediction strategies and their frequencies. Whether the constrained optimum has
 615 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
 616 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
 617 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
 618 interest frequency remain uniquely determined.

619 This equilibrium also allows a direct comparison with the expected accuracy of
 620 the AI. Using Y^{AI} itself without aggregation as the prediction gives

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S50})$$

621 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
 622 fore, because L_e is the minimum individual expected squared prediction error within
 623 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S51})$$

624 Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S52})$$

625 Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S53})$$

626 The collective prediction is the average of individual revised predictions. By
 627 convexity of the squared error,

$$\mathbb{E} [(Y - \hat{Y}^{re})^2] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S54})$$

Therefore,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] < \mathbb{E}[(Y - Y^{AI})^2]. \quad (\text{S55})$$

Because the collective accuracy and AI accuracy use the same normalization by $\mathbb{E}[(Y - \alpha_0)^2]$, the stationary collective accuracy under the niche-expert structure is strictly greater than the expected accuracy of the AI prediction.

Although the niche-expert structure improves collective prediction relative to the AI, the stationary collective accuracy does not reach one when $\tau > 0$. For a nonzero factor i , the residual coefficient of the stationary collective prediction can be written as

$$\delta_i^* = \alpha_i \{1 - \bar{\beta}^* - \rho_i^*(1 - \beta_i^*)\} = \alpha_i \sum_{e \neq i} \rho_e^*(1 - \beta_e^*). \quad (\text{S56})$$

At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore, for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S57})$$

Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error. Hence,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] > 0, \quad (\text{S58})$$

and consequently the stationary collective accuracy satisfies $R^* < 1$.

3.2 Chatbot AI

3.2.1 Feedback

With Chatbot AI, the effect of stochastic error on the collective prediction depends on n_i , requiring the additional assumption that $n_i \rightarrow \infty$ for every interest for the stochastic error to vanish. However, Under this assumption, the loss of interest diversity cannot occur, and therefore the stochastic breakdown of collective accuracy observed in finite populations cannot be analytically captured. Instead, by assuming that each interest is represented by infinitely many players, we can show that the expected collective prediction error monotonically decreases until perfect accuracy is achieved. Because finite population dynamics are more relevant to the empirical populations considered in this study, we retain finite population simulations for the main result.

3.2.2 Niche-expert

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S59})$$

657 After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its
 658 unconstrained stationary point is

$$\tilde{\beta}_e = \frac{2\alpha_0 b_e}{2b_e^2 + 2\tau^2}. \quad (\text{S60})$$

659 Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$,
 660 and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Because we
 661 assumes AI model with moderate accuracy, b_e and τ is relatively smaller than α_0
 662 which makes $\tilde{\beta}_e > 1$ when $\alpha_0 b_e > 0$ and $\tilde{\beta}_e < 0$ when $\alpha_0 b_e < 0$. These boundary
 663 cases lead to extreme reliance on AI or complete exclusion of AI which both hinders
 664 the emergence of CI. Only when the prediction task is centered near zero and AI is
 665 relatively inaccurate ($\alpha_0 b_e < b_e^2 + \tau^2$), and $\alpha_0 b_e > 0$, the internal optimum β_e^* can be
 666 exist.

667 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} b_0. \quad (\text{S61})$$

668 Substituting this into the expected payoff gives a strictly concave function of β , with
 669 the unique optimum

$$\beta_0^* = 0. \quad (\text{S62})$$

670 **Supplementary Figure**

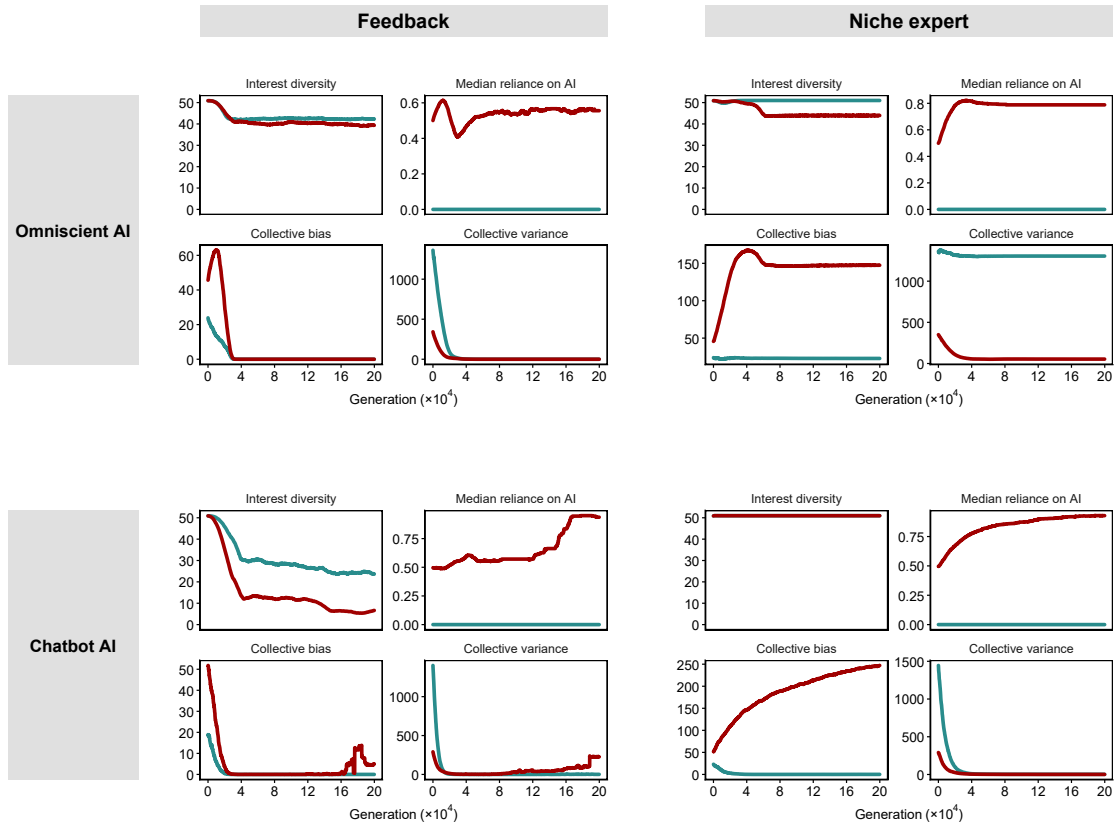


Figure S1: **Caption.** Text.

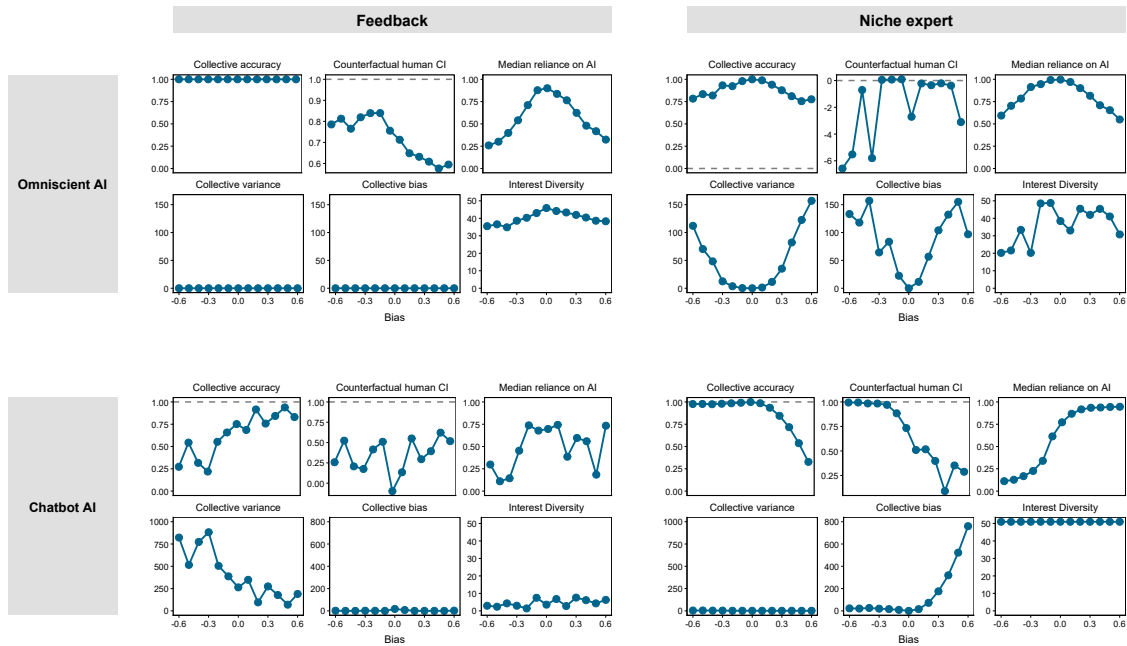


Figure S2: Caption. Text.

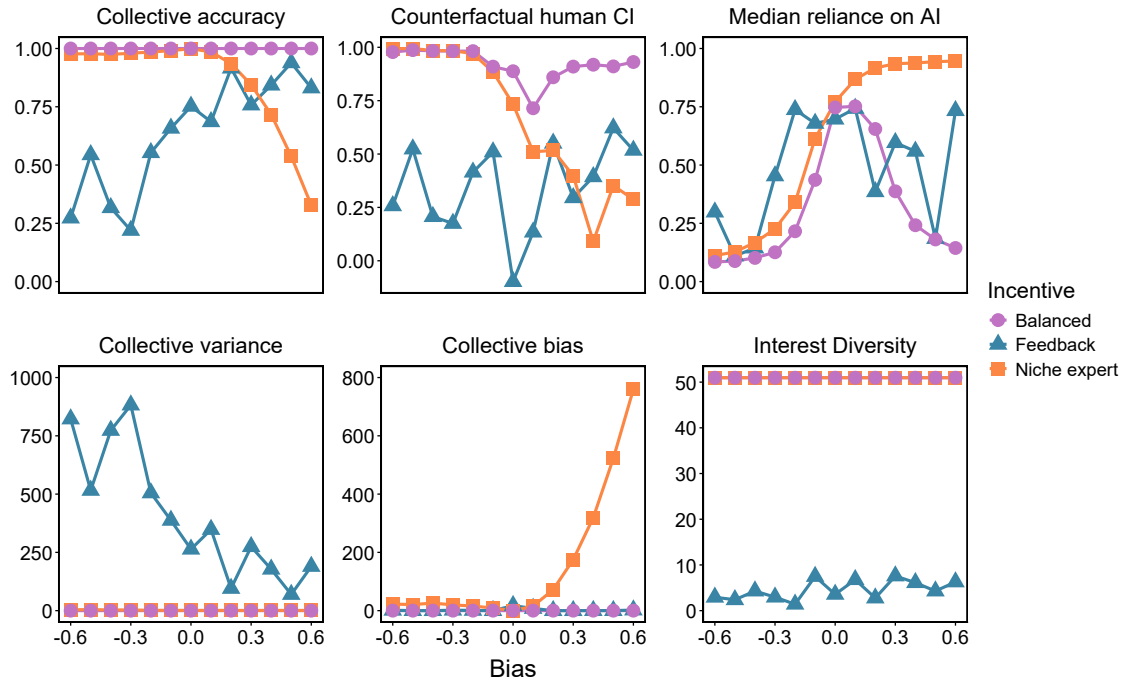


Figure S3: **Caption.** Text.

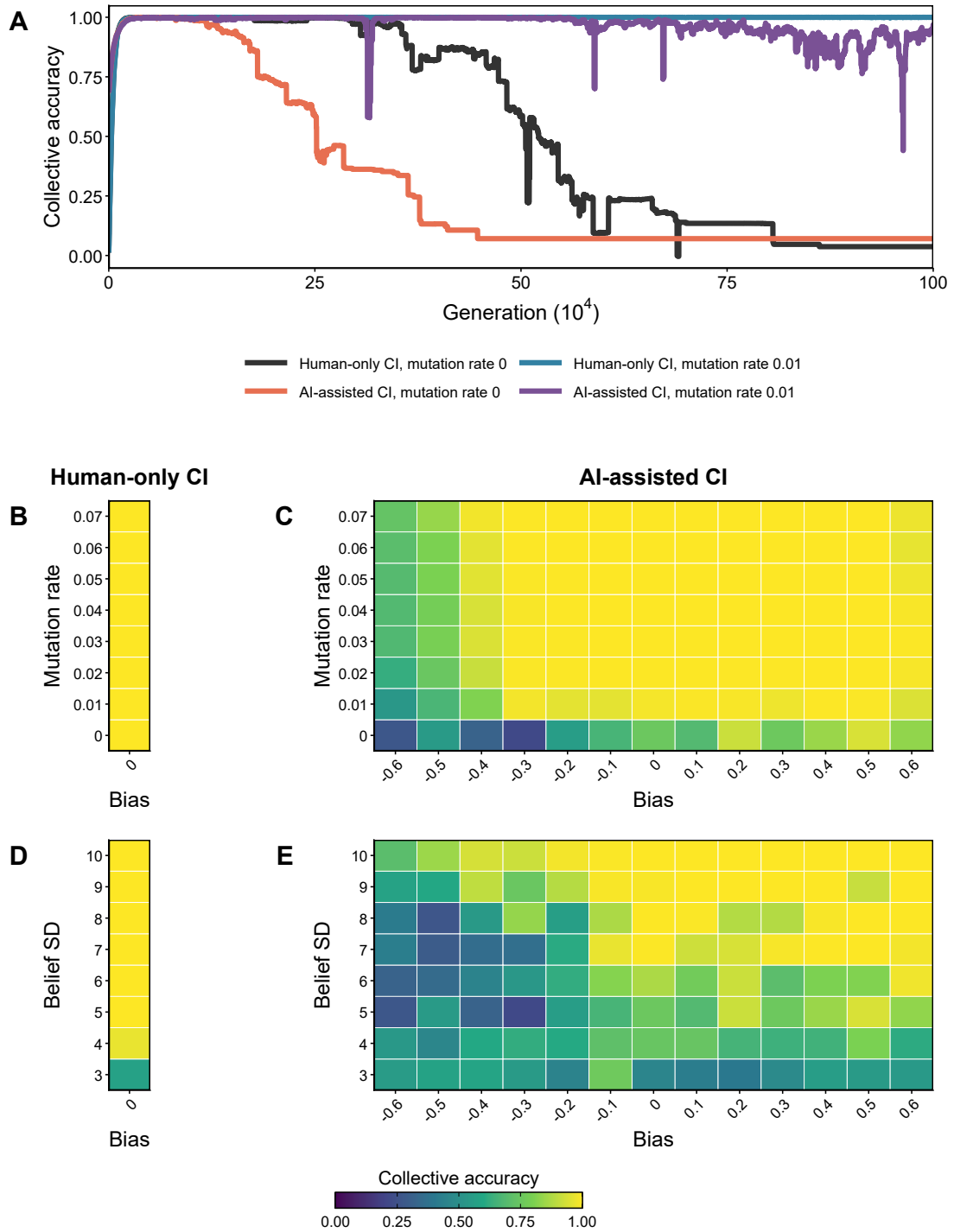


Figure S4: **Caption.** Text.

References

- [1] Francis Galton. Vox populi, 1907.
- [2] Anita Williams Woolley, Christopher F Chabris, Alex Pentland, Nada Hashmi, and Thomas W Malone. Evidence for a collective intelligence factor in the performance of human groups. *science*, 330(6004):686–688, 2010.
- [3] Ilan Yaniv and Maxim Milyavsky. Using advice from multiple sources to revise and improve judgments. *Organizational Behavior and Human Decision Processes*, 103(1):104–120, 2007.
- [4] Cars Hommes, Joep Sonnemans, Jan Tuinstra, and Henk Van de Velden. Coordination of expectations in asset pricing experiments. *The Review of Financial Studies*, 18(3):955–980, 2005.
- [5] Irving Lorge, David Fox, Joel Davitz, and Marlin Brenner. A survey of studies contrasting the quality of group performance and individual performance, 1920–1957. *Psychological bulletin*, 55(6):337, 1958.
- [6] Tabea Dreyer, Amir Haluts, Amos Korman, Nir Gov, Ehud Fonio, and Ofer Feinerman. Comparing cooperative geometric puzzle solving in ants versus humans. *Proceedings of the National Academy of Sciences*, 122(1):e2414274121, 2025.
- [7] Andrew M Simons. Many wrongs: the advantage of group navigation. *Trends in ecology & evolution*, 19(9):453–455, 2004.
- [8] Andrew Berdahl, Colin J Torney, Christos C Ioannou, Jolyon J Faria, and Iain D Couzin. Emergent sensing of complex environments by mobile animal groups. *Science*, 339(6119):574–576, 2013.
- [9] Christoph Riedl, Young Ji Kim, Pranav Gupta, Thomas W Malone, and Anita Williams Woolley. Quantifying collective intelligence in human groups. *Proceedings of the National Academy of Sciences*, 118(21):e2005737118, 2021.
- [10] Leo Breiman. Bagging predictors. *Machine learning*, 24(2):123–140, 1996.
- [11] Leo Breiman. Random forests. *Machine learning*, 45(1):5–32, 2001.
- [12] Richard P Mann and Dirk Helbing. Optimal incentives for collective intelligence. *Proceedings of the National Academy of Sciences*, 114(20):5077–5082, 2017.
- [13] Guocheng Wang, Qi Su, Long Wang, and Joshua B Plotkin. Individual incentives that promote collective intelligence. *Proceedings of the National Academy of Sciences*, 122(51):e2516535122, 2025.

- [14] Justin Werfel. Fluid thinking about collective intelligence. *Nature Machine Intelligence*, 8(4):506–516, 2026.
- [15] Hao Cui and Taha Yasseri. AI-enhanced collective intelligence. *Patterns*, 5(11):101074, 2024.
- [16] Chris Lu, Cong Lu, Robert Tjarko Lange, Yutaro Yamada, Shengran Hu, Jakob Foerster, David Ha, and Jeff Clune. Towards end-to-end automation of AI research. *Nature*, 651(8107):914–919, 2026.
- [17] Anil R Doshi and Oliver P Hauser. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science advances*, 10(28):eadn5290, 2024.
- [18] Sha Sajadieh, Loredana Fattorini, Raymond Perrault, Yolanda Gil, Vanessa Parli, Lapo Santarasci, Juan Pava, Nestor Maslej, Russ Altman, Erik Brynjolfsson, et al. Artificial intelligence index report 2026. *arXiv preprint arXiv:2606.15708*, 2026.
- [19] Michelle Vaccaro, Abdullah Almaatouq, and Thomas Malone. When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8(12):2293–2303, 2024.
- [20] Jason W Burton, Ezequiel Lopez-Lopez, Shahar Hechtlinger, Zoe Rahwan, Samuel Aeschbach, Michiel A Bakker, Joshua A Becker, Aleks Berditchevskaia, Julian Berger, Levin Brinkmann, et al. How large language models can reshape collective intelligence. *Nature human behaviour*, 8(9):1643–1655, 2024.
- [21] Lennart Meincke, Gideon Nave, and Christian Terwiesch. ChatGPT decreases idea diversity in brainstorming. *Nature human behaviour*, 9(6):1107–1109, 2025.
- [22] Andreas Fügener, Jörn Grahl, Alok Gupta, and Wolfgang Ketter. Will humans-in-the-loop become borgs? merits and pitfalls of working with AI. *MIS quarterly*, 45(3):1527–1556, 2021.
- [23] Jan Lorenz, Heiko Rauhut, Frank Schweitzer, and Dirk Helbing. How social influence can undermine the wisdom of crowd effect. *Proceedings of the national academy of sciences*, 108(22):9020–9025, 2011.
- [24] Lucy M Aplin, Damien R Farine, Richard P Mann, and Ben C Sheldon. Individual-level personality influences social foraging and collective behaviour in wild birds. *Proceedings of the Royal Society B: Biological Sciences*, 281(1789):20141016, 2014.
- [25] Lu Hong and Scott E Page. Groups of diverse problem solvers can outperform groups of high-ability problem solvers. *Proceedings of the National Academy of Sciences*, 101(46):16385–16389, 2004.

- 740 [26] Isabelle Bousquette. Why some companies say AI ‘token-
 741 maxxng’ is key to survival. [https://www.wsj.com/cio-journal/](https://www.wsj.com/cio-journal/why-some-companies-say-ai-tokenmaxxing-is-key-to-survival-e699a128)
 742 [why-some-companies-say-ai-tokenmaxxing-is-key-to-survival-e699a128](https://www.wsj.com/cio-journal/why-some-companies-say-ai-tokenmaxxing-is-key-to-survival-e699a128),
 743 April 2026. The Wall Street Journal. Accessed: 2026-07-07.
- 744 [27] Tim Keary. Is the cult of ‘tokenmaxxing’ just another fad or the
 745 new normal? [https://www.forbes.com/sites/timkeary/2026/04/13/](https://www.forbes.com/sites/timkeary/2026/04/13/is-the-cult-of-tokenmaxxingjust-another-fad-or-the-new-normal/)
 746 [is-the-cult-of-tokenmaxxingjust-another-fad-or-the-new-normal/](https://www.forbes.com/sites/timkeary/2026/04/13/is-the-cult-of-tokenmaxxingjust-another-fad-or-the-new-normal/),
 747 April 2026. Forbes. Accessed: 2026-07-07.
- 748 [28] Dalmeet Singh Chawla. Researchers who use hallucinated references to face
 749 arXiv ban. *Nature*, 653(8116):988–989, 2026.
- 750 [29] H Holden Thorp. Chatgpt is fun, but not an author. *Science*, 379(6630):313–313,
 751 2023.
- 752 [30] Lu Hong, Scott E Page, and Maria Riolo. Incentives, information, and emergent
 753 collective accuracy. *Managerial and Decision Economics*, 33(5-6):323–334, 2012.
- 754 [31] Qianyu Hao, Fengli Xu, Yong Li, and James Evans. Artificial intelligence tools
 755 expand scientists’ impact but contract science’s focus. *Nature*, 649(8099):1237–
 756 1243, 2026.
- 757 [32] Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and
 758 Rei Mariman. Generative AI without guardrails can harm learning: Evidence
 759 from high school mathematics. *Proceedings of the National Academy of Sciences*,
 760 122(26):e2422633122, 2025.